LAB ASSIGNMENT – 4
DATE – 09.02.2026

NAME : Shreya Ghoshal
REGISTRATION NO. : AU/2023/0009045

ROLL NO. : UG/02/BTCSECSF/2023/003

SUBJECT : Unstructured Database Lab
SEC: E

Schema

{ roll_no, name, department, semester, city, marks}

Initial Data

RollNo.	Name	Department	Semester	City	Marks
1	Amit	CSE	4	Kolkata	78
2	Neha	IT	4	Delhi	65
3	Ravi	CSE	5	Mumbai	82
4	Pooja	ECE	4	Chennai	70
5	Karan	IT	5	Kolkata	55
6	Sneha	CSE	4	Delhi	88

Database Name: **collegeDB_'YourRollNo.'**

Collection Name: **students_' YourRollNo.'**

Lab Questions:

Q1. Display all student records using aggregation.

```
college_YourRollNo> db.studentList_YourRollNo.aggregate([
...   { $match: {} }
... ])
[
  {
    _id: ObjectId('69897fbe102ccdcd79628ca0'),
    roll_no: 1,
    name: 'Amit',
    department: 'CSE',
    semester: 4,
    city: 'Kolkata',
    marks: 78
  },
  {
    _id: ObjectId('69897fbe102ccdcd79628ca1'),
    roll_no: 2,
    name: 'Neha',
    department: 'IT',
    semester: 4,
    city: 'Delhi',
    marks: 65
  },
  {
    _id: ObjectId('69897fbe102ccdcd79628ca2'),
    roll_no: 3,
    name: 'Ravi',
    department: 'CSE',
    semester: 5,
    city: 'Mumbai',
    marks: 82
  },
  {
    _id: ObjectId('69897fbe102ccdcd79628ca3'),
    roll_no: 4,
    name: 'Pooja',
    department: 'ECE',
    semester: 4,
    city: 'Chennai',
    marks: 70
  },
  {
    _id: ObjectId('69897fbe102ccdcd79628ca4'),
    roll_no: 5,
    name: 'Karan',
    department: 'IT',
    semester: 5,
    city: 'Kolkata',
    marks: 55
  },
  {
    _id: ObjectId('69897fbe102ccdcd79628ca5'),
    roll_no: 6,
    name: 'Sneha',
    department: 'CSE',
    semester: 4,
    city: 'Delhi',
    marks: 88
  }
]
```

Q2. Display students belonging to CSE department.

```
college_YourRollNo> db.studentList_YourRollNo.aggregate([
...   { $match: { department: "CSE" } }
... ])
[
  {
    _id: ObjectId('69897fbe102ccdcd79628ca0'),
    roll_no: 1,
    name: 'Amit',
    department: 'CSE',
    semester: 4,
    city: 'Kolkata',
    marks: 78
  },
  {
    _id: ObjectId('69897fbe102ccdcd79628ca2'),
    roll_no: 3,
    name: 'Ravi',
    department: 'CSE',
    semester: 5,
    city: 'Mumbai',
    marks: 82
  },
  {
    _id: ObjectId('69897fbe102ccdcd79628ca5'),
    roll_no: 6,
    name: 'Sneha',
    department: 'CSE',
    semester: 4,
    city: 'Delhi',
    marks: 88
  }
]
```

Q3. Display students of semester 4.

```
college_YourRollNo> db.studentList_YourRollNo.aggregate([
...   { $match: { semester: 4 } }
... ])
[
  {
    _id: ObjectId('69897fbe102ccdcd79628ca0'),
    roll_no: 1,
    name: 'Amit',
    department: 'CSE',
    semester: 4,
    city: 'Kolkata',
    marks: 78
  },
  {
    _id: ObjectId('69897fbe102ccdcd79628ca1'),
    roll_no: 2,
    name: 'Neha',
    department: 'IT',
    semester: 4,
    city: 'Delhi',
    marks: 65
  },
  {
    _id: ObjectId('69897fbe102ccdcd79628ca3'),
    roll_no: 4,
    name: 'Pooja',
    department: 'ECE',
    semester: 4,
    city: 'Chennai',
    marks: 70
  },
  {
    _id: ObjectId('69897fbe102ccdcd79628ca5'),
    roll_no: 6,
    name: 'Sneha',
    department: 'CSE',
    semester: 4,
    city: 'Delhi',
    marks: 88
  }
]
```

Q4. Display only name and marks of all students.

```
college_YourRollNo> db.studentList_YourRollNo.aggregate([
...   { $project: { _id: 0, name: 1, marks: 1 } }
... ])
[
  { name: 'Amit', marks: 78 },
  { name: 'Neha', marks: 65 },
  { name: 'Ravi', marks: 82 },
  { name: 'Pooja', marks: 70 },
  { name: 'Karan', marks: 55 },
  { name: 'Sneha', marks: 88 }
]
```

Q5. Find the average marks of all students.

```
college_YourRollNo> db.studentList_YourRollNo.aggregate([
...   {
...     $group: {
...       _id: null,
...       averageMarks: { $avg: "$marks" }
...     }
...   }
... ])
[ { _id: null, averageMarks: 73 } ]
```

Q6. Find total marks scored department-wise.

```
college_YourRollNo> db.studentList_YourRollNo.aggregate([
...   {
...     $group: {
...       _id: "$department",
...       totalMarks: { $sum: "$marks" }
...     }
...   }
... ])
[
  { _id: 'ECE', totalMarks: 70 },
  { _id: 'CSE', totalMarks: 248 },
  { _id: 'IT', totalMarks: 120 }
]
```

Q7.Find average marks department wise.

```
college_YourRollNo> db.studentList_YourRollNo.aggregate([
...   {
...     $group: {
...       _id: "$department",
...       averageMarks: { $avg: "$marks" }
...     }
...   }
... ])
[
  { _id: 'ECE', averageMarks: 70 },
  { _id: 'CSE', averageMarks: 82.66666666666667 },
  { _id: 'IT', averageMarks: 60 }
]
```

Q8. Find maximum marks scored in each department.

```
college_YourRollNo> db.studentList_YourRollNo.aggregate([
...   {
...     $group: {
...       _id: "$department",
...       maxMarks: { $max: "$marks" }
...     }
...   }
... ])
[
  { _id: 'IT', maxMarks: 65 },
  { _id: 'CSE', maxMarks: 88 },
  { _id: 'ECE', maxMarks: 70 }
]
```

Q9. Count number of students in each department.

```
college_YourRollNo> db.studentList_YourRollNo.aggregate([
...   {
...     $group: {
...       _id: "$department",
...       studentCount: { $sum: 1 }
...     }
...   }
... ])
[
  { _id: 'ECE', studentCount: 1 },
  { _id: 'CSE', studentCount: 3 },
  { _id: 'IT', studentCount: 2 }
]
```

Q10. Display departments having average marks greater than 70.

```
college_YourRollNo> db.studentList_YourRollNo.aggregate([
...   {
...     $group: {
...       _id: "$department",
...       averageMarks: { $avg: "$marks" }
...     }
...   },
...   {
...     $match: { averageMarks: { $gt: 70 } }
...   }
... ])
[ { _id: 'CSE', averageMarks: 82.66666666666667 } ]
```

Q11.Sort students by marks in descending order.

```
college_YourRollNo> db.studentList_YourRollNo.aggregate([
...   { $sort: { marks: -1 } }
... ])
[
  {
    _id: ObjectId('69897fbe102ccdcd79628ca5'),
    roll_no: 6,
    name: 'Sneha',
    department: 'CSE',
    semester: 4,
    city: 'Delhi',
    marks: 88
  },
  {
    _id: ObjectId('69897fbe102ccdcd79628ca2'),
    roll_no: 3,
    name: 'Ravi',
    department: 'CSE',
    semester: 5,
    city: 'Mumbai',
    marks: 82
  },
  {
    _id: ObjectId('69897fbe102ccdcd79628ca0'),
    roll_no: 1,
    name: 'Amit',
    department: 'CSE',
    semester: 4,
    city: 'Kolkata',
    marks: 78
  },
  {
    _id: ObjectId('69897fbe102ccdcd79628ca3'),
    roll_no: 4,
    name: 'Pooja',
    department: 'ECE',
    semester: 4,
    city: 'Chennai',
    marks: 70
  },
  {
    _id: ObjectId('69897fbe102ccdcd79628ca1'),
    roll_no: 2,
    name: 'Neha',
    department: 'IT',
    semester: 4,
    city: 'Delhi',
    marks: 65
  },
  {
    _id: ObjectId('69897fbe102ccdcd79628ca4'),
    roll_no: 5,
    name: 'Karan',
    department: 'IT',
    semester: 5,
    city: 'Kolkata',
    marks: 55
  }
]
```

Q12. Display departments having average marks greater than 70.

```
college_YourRollNo> db.studentList_YourRollNo.aggregate([
...   { $sort: { marks: -1 } },
...   { $limit: 3 }
... ])
[
  {
    _id: ObjectId('69897fbe102ccdcd79628ca5'),
    roll_no: 6,
    name: 'Sneha',
    department: 'CSE',
    semester: 4,
    city: 'Delhi',
    marks: 88
  },
  {
    _id: ObjectId('69897fbe102ccdcd79628ca2'),
    roll_no: 3,
    name: 'Ravi',
    department: 'CSE',
    semester: 5,
    city: 'Mumbai',
    marks: 82
  },
  {
    _id: ObjectId('69897fbe102ccdcd79628ca0'),
    roll_no: 1,
    name: 'Amit',
    department: 'CSE',
    semester: 4,
    city: 'Kolkata',
    marks: 78
  }
]
```

Q13. Display department-wise average marks stored in descending order.

```
college_YourRollNo> db.studentList_YourRollNo.aggregate([
...   {
...     $group: {
...       _id: "$department",
...       averageMarks: { $avg: "$marks" }
...     }
...   },
...   { $sort: { averageMarks: -1 } }
... ])
[
  { _id: 'CSE', averageMarks: 82.66666666666667 },
  { _id: 'ECE', averageMarks: 70 },
  { _id: 'IT', averageMarks: 60 }
]
```

Q14. Display students belonging to Delhi or Kolkata.

```
college_YourRollNo> db.studentList_YourRollNo.aggregate([
...   {
...     $match: {
...       city: { $in: ["Delhi", "Kolkata"] }
...     }
...   }
... ])
[
  {
    _id: ObjectId('69897fbe102ccdcd79628ca0'),
    roll_no: 1,
    name: 'Amit',
    department: 'CSE',
    semester: 4,
    city: 'Kolkata',
    marks: 78
  },
  {
    _id: ObjectId('69897fbe102ccdcd79628ca1'),
    roll_no: 2,
    name: 'Neha',
    department: 'IT',
    semester: 4,
    city: 'Delhi',
    marks: 65
  },
  {
    _id: ObjectId('69897fbe102ccdcd79628ca4'),
    roll_no: 5,
    name: 'Karan',
    department: 'IT',
    semester: 5,
    city: 'Kolkata',
    marks: 55
  },
  {
    _id: ObjectId('69897fbe102ccdcd79628ca5'),
    roll_no: 6,
    name: 'Sneha',
    department: 'CSE',
    semester: 4,
    city: 'Delhi',
    marks: 88
  }
]
```